

# Chapter 26: Vision and Optical Instruments

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PHY201  
Lecture 26



# Part I

## The Human Eye

# Outline of Part I: The Human Eye

Introduction

Anatomy of the Eye

Accommodation and Near/Far Point

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Anatomy of the Eye

Accommodation and Near/Far Point

# Vision and Optical Instruments

The eye is the most sophisticated optical instrument ever developed — a self-focusing, self-adjusting, self-repairing lens system that can detect a single photon.

## Central Question of This Chapter

- ▶ How do optical systems collect light and form images?
- ▶ How does the eye fail — and how do we correct it?
- ▶ How do microscopes and telescopes extend human vision?
- ▶ Why are all real optical systems imperfect?

## Connecting Thread

Each topic in this chapter answers the same question:

*How does an optical system form an image of an object, and what are its limits?*

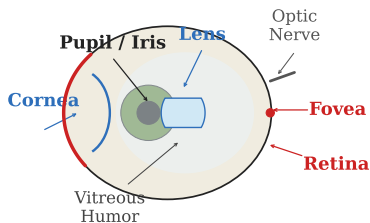
# Outline of Part I: The Human Eye

Introduction

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# Anatomy of the Eye



## Key Structures

- ▶ **Cornea** — primary refracting surface ( $\sim 2/3$  of total power)
- ▶ **Lens** — adjustable, provides fine focus
- ▶ **Iris/Pupil** — controls light intensity
- ▶ **Retina** — light-sensitive detector
- ▶ **Fovea** — highest acuity region
- ▶ **Optic nerve** — signal pathway to brain

# The Eye as a Converging Lens System

## Thin-Lens Equation Applied to the Eye

The image distance  $d_i$  is **fixed** ( $\approx 2.0$  cm):

$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} = P}$$

$P = 1/f$ : power in **diopters** (D)

Since  $d_i$  is fixed, the eye adjusts  $f$  (via  $P$ ) to focus objects at different distances.

## Power of the Eye

- ▶ Total at rest:  $\approx 60$  D
- ▶ Cornea:  $\approx 43$  D (fixed)
- ▶ Lens:  $\approx 17$  D (adjustable)
- ▶ Maximum:  $\approx 67$  D

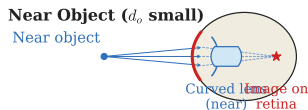
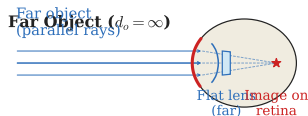
# Outline of Part I: The Human Eye

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# Accommodation: Adjusting Focus



## Accommodation

The ciliary muscles **change the shape** of the lens, altering its focal length.

- ▶ **Far object:** ciliary muscles relax, lens flattens,  $f$  increases ( $P$  decreases)
- ▶ **Near object:** ciliary muscles contract, lens rounds,  $f$  decreases ( $P$  increases)

The **image must always land on the retina**. Since  $d_i$  is fixed,  $f$  must change when  $d_o$  changes.

# Near Point and Far Point

## Far Point

The **farthest** distance at which the eye can form a clear image.

A normal (emmetropic) eye: far point  
 $= \infty$

( $d_o = \infty \Rightarrow P = 1/d_i$ , minimum power)

## Near Point

The **closest** distance at which the eye can form a clear image.

Normal young adult: near point  
 $\approx 25 \text{ cm}$

( $d_o = 25 \text{ cm} \Rightarrow P$  maximum)

## Accommodation Range

$$\begin{aligned}\Delta P &= P_{\max} - P_{\min} \\ &= \frac{1}{d_{o,\text{near}}} + \frac{1}{d_i} - \frac{1}{d_i} \\ &= \frac{1}{d_{o,\text{near}}}\end{aligned}$$

As we age, the lens hardens and accommodation range decreases — this is **presbyopia**.

# The Inverted Retinal Image

The lens of the eye forms a **real, inverted** image on the retina — just as any converging lens forms an inverted real image when  $d_o > f$ .

## Common Misconception

- ▶ Students often assume we see the world “right-side up” because the lens rights the image.
- ▶ In fact, the retinal image is **inverted** and **reduced**.
- ▶ The brain interprets the neural signals from the retina and constructs an upright perception.

## Key Insight

**Perception** is constructed by the brain, not recorded passively by the eye.

The eye is the *detector*; the brain is the *image processor*.

## Example: Power of the Eye — (cp2e\_ch26\_ex1)

### Power of the Eye for a Distant Object

Calculate the power of the eye when viewing an object at infinity. Assume  $d_i = 2.00$  cm.

$$\frac{1}{d_o} + \frac{1}{d_i} = P$$

$$d_o = \infty \Rightarrow \frac{1}{d_o} = 0$$

$$P = \frac{1}{d_i} = \frac{1}{0.0200 \text{ m}}$$

$$P = 50.0 \text{ D}$$

At rest (far point =  $\infty$ ), the eye operates at minimum power: 50.0 D.

The total power  $\approx 60$  D includes a corneal contribution of  $\approx 43$  D — but the simple model treats the eye as a single thin lens.

- What happens to  $P$  as  $d_o$  decreases?
- Why can't  $d_i$  change to refocus?

## Example: Near-Point Power — (cp2e\_ch26\_ex2)

### Maximum Power of the Eye

An eye has a near point of 25.0 cm and  $d_i = 2.00$  cm. What is its maximum power?

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \frac{1}{0.250 \text{ m}} + \frac{1}{0.0200 \text{ m}}$$

$$P = 4.00 \text{ D} + 50.0 \text{ D}$$

$$P = 54.0 \text{ D}$$

Accommodation range:

$$54.0 \text{ D} - 50.0 \text{ D} = 4.0 \text{ D}.$$

This is the extra power the lens muscles must supply when focusing at the near point.

- ▶ Is  $P_{\max} > P_{\min}$ ? ✓
- ▶ Why does the near point move farther away with age?

## Check Your Understanding

An object is moved closer to a normal eye. Does the power of the eye increase or decrease? Does the focal length increase or decrease?

Come to lecture to see the answer.

## Part II

# Vision Correction

# Outline of Part II: Vision Correction

Myopia and Hyperopia

Presbyopia and Laser Correction

# Outline of Part II: Vision Correction

Myopia and Hyperopia

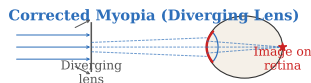
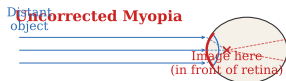
Presbyopia and Laser Correction

## Vision Defects: When Image Formation Fails

When the eye cannot bring an image to focus on the retina — because the eyeball is the wrong size or the lens cannot adjust enough — we need a **corrective lens**.

| Condition  | Cause                                               | Correction                   |
|------------|-----------------------------------------------------|------------------------------|
| Myopia     | Eye too long; image falls <i>in front of</i> retina | Diverging (concave) lens     |
| Hyperopia  | Eye too short; image falls <i>behind</i> retina     | Converging (convex) lens     |
| Presbyopia | Reduced accommodation range                         | Reading glasses (converging) |

# Myopia (Nearsightedness)



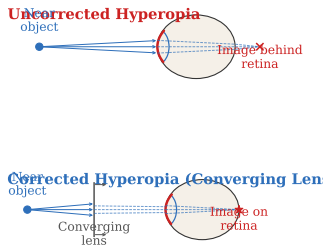
## Myopia

- ▶ Far point is *closer* than infinity
- ▶ Distant objects focus in front of retina
- ▶ Cannot see far objects clearly

## Correction

A **diverging lens** makes incoming parallel rays appear to come from the far point — placing the image where the myopic eye *can* focus it.

# Hyperopia (Farsightedness)



## Hyperopia

- ▶ Near point is *farther* than normal
- ▶ Near objects focus behind retina
- ▶ Cannot see near objects clearly

## Correction

A **converging lens** pre-converges the rays from a near object, making them appear to come from the eye's near point or beyond.

## Finding the Corrective Lens Power

### Strategy for Vision Correction

A corrective lens creates a **virtual image** at the eye's far point (myopia) or near point (hyperopia), so the eye can focus it.

The object is at the *desired viewing distance*; the image is at the *limit of the defective eye*.

### Myopia: Correcting Lens

Object at  $d_o = \infty$  (distant) must form image at the far point  $d_i = -|\text{far point}|$  (virtual, same side as object).

$$P_{\text{lens}} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{\infty} - \frac{1}{|\text{far pt}|}$$

### Hyperopia: Correcting Lens

Object at  $d_o = 25 \text{ cm}$  (near point) must form image at the eye's near point  $d_i = -|\text{near pt}|$  (virtual).

$$P_{\text{lens}} = \frac{1}{d_o} - \frac{1}{|\text{near pt}|}$$

## Example: Correcting Myopia — (cp2e\_ch26\_ex3)

### Correcting Myopia

A person has a far point of 80.0 cm. What power corrective lens do they need to see distant objects clearly?

$$d_o = \infty, \quad d_i = -0.800 \text{ m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = 0 + \frac{1}{-0.800 \text{ m}}$$

$$P = -1.25 \text{ D}$$

Negative power  $\Rightarrow$  **diverging lens** — correct for myopia.

The virtual image at  $-80.0 \text{ cm}$  is at the person's far point; the myopic eye can now focus it.

- Why is  $d_i$  negative here?
- Does a more negative  $P$  correct more or less severe myopia?

## Example: Correcting Hyperopia — (cp2e\_ch26\_ex4)

### Correcting Hyperopia

A farsighted person has a near point of 60.0 cm. What power lens allows them to read at 25.0 cm?

$$d_o = 0.250 \text{ m}, \quad d_i = -0.600 \text{ m}$$

$$P = \frac{1}{d_o} + \frac{1}{d_i}$$

$$P = \frac{1}{0.250 \text{ m}} + \frac{1}{-0.600 \text{ m}}$$

$$P = 4.00 \text{ D} - 1.67 \text{ D}$$

$$P = 2.33 \text{ D}$$

Positive power  $\Rightarrow$  **converging lens** — correct for hyperopia.

The virtual image at  $-60.0 \text{ cm}$  is at the eye's near point.

- Is the image real or virtual? Why?
- Does the corrective lens replace the eye's focusing or assist it?

# Outline of Part II: Vision Correction

Myopia and Hyperopia

Presbyopia and Laser Correction

# Presbyopia and Laser Surgery

## Presbyopia

- ▶ Age-related reduction in accommodation
- ▶ Lens hardens and cannot round sufficiently
- ▶ Near point recedes beyond comfortable reading distance
- ▶ Corrected with **reading glasses** (converging)
- ▶ Not the same as hyperopia — the eye is the right size, but the lens cannot deform

## Laser Vision Correction

- ▶ LASIK reshapes the **cornea**, permanently changing its curvature
- ▶ Changes optical power at the source
- ▶ Does *not* strengthen muscles or restore accommodation
- ▶ Most effective for myopia and mild astigmatism

Laser surgery changes the *geometry* of the cornea — it is optical engineering applied to biological tissue.

## Check Your Understanding

A patient's far point is 50 cm. Is this person myopic or hyperopic? What sign of corrective lens power do they need?

Come to lecture to see the answer.

## Part III

# Color and Color Vision

# Outline of Part III: Color and Color Vision

The Visible Spectrum

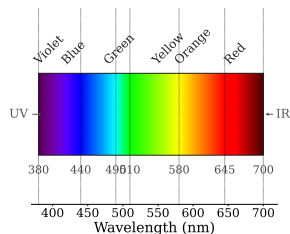
Rods, Cones, and Color Mixing

# Outline of Part III: Color and Color Vision

The Visible Spectrum

Rods, Cones, and Color Mixing

# The Visible Spectrum



## Visible Range

- ▶  $\lambda \approx 380 \text{ nm}$  (violet) to  $700 \text{ nm}$  (red)
- ▶ A narrow slice of the full EM spectrum
- ▶ **Color is a perception**, not a physical property of light
- ▶ Same wavelength can appear different colors under different contexts

## Key Distinction

**Physical:** spectral distribution of incoming light  
interpretation of receptor signals

**Perceived:** brain's

# Outline of Part III: Color and Color Vision

The Visible Spectrum

Rods, Cones, and Color Mixing

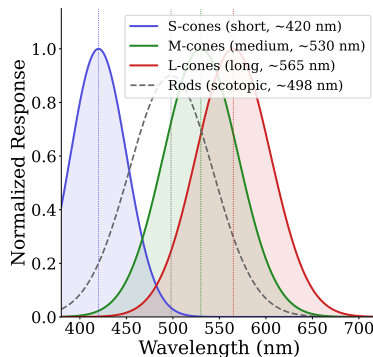
# Rods and Cones

## Rods

- ▶ ~120 million in each eye
- ▶ Sensitive to low light levels (night vision)
- ▶ No color discrimination
- ▶ Absent from fovea; concentrated in periphery

## Cones

- ▶ ~6 million in each eye
- ▶ Require bright light (day vision)
- ▶ Color sensitive: three types (S, M, L)
- ▶ Concentrated in fovea



# Trichromatic Theory of Color Vision

Color perception arises from the **ratio of responses** of three cone types: S (short,  $\sim 420$  nm), M (medium,  $\sim 530$  nm), L (long,  $\sim 560$  nm).

## Metamerism

Two *physically different* spectra can produce the *identical perception* if they stimulate the three cone types in the same ratio.

Example: pure yellow light ( $\sim 580$  nm) and a red-green mixture can appear identical — even though the spectra are completely different.

## Practical Applications

- ▶ TV/computer displays use only 3 colors (RGB)
- ▶ Printing uses CMYK
- ▶ Human eye cannot distinguish metamers
- ▶ Camera sensors mimic trichromatic detection

## Additive vs. Subtractive Color Mixing

### Additive (RGB)

Screens, projectors

Emit light

$R + G + B \rightarrow \text{white}$

$R + G = \text{Yellow}$

**Why different?** Screens *emit* light; adding colors adds photons. Pigments *remove* wavelengths; mixing removes more.

### Subtractive (CMY)

Paints, inks, pigments

Absorb light

$C + M + Y \rightarrow \text{black}$

C absorbs Red

Cyan = complement of Red  
Magenta = complement of Green  
Yellow = complement of Blue

# Color Blindness

Color blindness results from the **absence or malfunction** of one or more cone types.

- ▶ **Deuteranopia:** missing M-cones (green); red-green confusion — most common (~5% of men)
- ▶ **Protanopia:** missing L-cones (red); red-green confusion
- ▶ **Tritanopia:** missing S-cones (blue); rare
- ▶ **Monochromacy:** only one cone type; rare

## Signal encoding view:

Three cone types → 3D color space.

Two cone types → 2D color space.

One cone type → 1D (no hue).

Color blindness = reduced dimensionality of color encoding.

Color blind individuals lose *one dimension* of the color signal; fewer colors can be distinguished.

## Check Your Understanding

Two different light sources produce the same color perception in a normal observer. Does this mean they emit the same wavelengths? Explain.

Come to lecture to see the answer.

## Part IV

# Microscopes

## Outline of Part IV: Microscopes

Simple Magnifier

The Compound Microscope

Resolution and Numerical Aperture

## Outline of Part IV: Microscopes

Simple Magnifier

The Compound Microscope

Resolution and Numerical Aperture

## The Simple Magnifier

A **simple magnifier** (magnifying glass) is a single converging lens used to view objects placed closer than its focal length, producing a **virtual, upright, enlarged** image.

### Angular Magnification

The magnifying glass increases the **angular size** of the object as seen by the eye.

$$M = \frac{\theta'}{\theta} \approx \frac{25 \text{ cm}}{f}$$

where 25 cm is the standard near-point distance and the image is at infinity ( $d_o = f$ ).

- ▶ Shorter  $f \Rightarrow$  larger  $M$
- ▶ A  $10\times$  lens has  $f \approx 2.5 \text{ cm}$
- ▶ Object placed at the focal point: image at  $\infty$ , eye relaxed
- ▶ Object inside  $f$ : virtual, upright, magnified

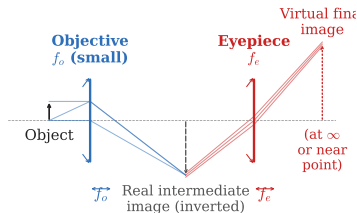
# Outline of Part IV: Microscopes

Simple Magnifier

The Compound Microscope

Resolution and Numerical Aperture

# The Compound Microscope



## Two-Lens System

- ▶ **Objective lens** ( $f_o$  small): creates a real, inverted, *enlarged* intermediate image
- ▶ **Eyepiece lens** ( $f_e$ ): acts as magnifying glass for the intermediate image
- ▶ **Final image**: virtual, inverted, very large

## Image Chain

Object  $\xrightarrow{\text{objective}}$  real intermediate image  
image  $\xrightarrow{\text{eyepiece}}$  virtual final image

# Magnification of the Compound Microscope

## Total Magnification

The total magnification is the **product** of objective and eyepiece magnifications:

$$m = m_o \cdot m_e$$

## Objective Magnification

$$m_o \approx -\frac{d_i}{f_o}$$

$d_i$  = image distance from objective  
(large)

Negative: intermediate image is  
inverted

## Eyepiece Magnification

**Example:** a microscope with  
 $m_o = 40\times$  objective and  $m_e = 10\times$   
eyepiece gives total  $m = 400\times$ .

## Bigger $\neq$ More Detailed

Magnification alone does not reveal  
finer detail. **Resolution** is what  
determines the smallest visible feature.

# Outline of Part IV: Microscopes

Simple Magnifier

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Resolution and Numerical Aperture

# Resolution vs. Magnification

## Empty Magnification

Magnifying beyond the resolution limit produces a **larger blur**.

Bigger image  $\neq$  more detail.

## Numerical Aperture

$$NA = n \sin \alpha$$

$n$ : index;  $\alpha$ : half-angle of light cone at objective

## Rayleigh Resolution Limit

$$\delta \approx \frac{0.61 \lambda}{NA}$$

- ▶ Smaller  $\delta$  = sharper image
- ▶ Short  $\lambda$  or large NA improves resolution
- ▶ Oil immersion:  $n > 1 \Rightarrow$  larger NA

## Example: Compound Microscope Magnification — (cp2e\_ch26\_ex5)

### Compound Microscope

$f_o = 5.00 \text{ mm}$ ,  $f_e = 25.0 \text{ mm}$ , image distance  $d_i = 15.0 \text{ cm}$ . Find  $m$ .

$$m_o \approx -\frac{d_i}{f_o} = -\frac{150 \text{ mm}}{5.00 \text{ mm}} = -30.0$$

$$m_e = \frac{250 \text{ mm}}{f_e} = \frac{250 \text{ mm}}{25.0 \text{ mm}} = 10.0$$

$$m = m_o \cdot m_e = (-30.0)(10.0)$$

$$m = -300$$

$|m| = 300\times$ ; negative sign  $\Rightarrow$  final image is **inverted**.

- ▶ Which lens contributes more magnification?
- ▶ What limits how much we can magnify?

## Check Your Understanding

A microscope is magnifying at  $400\times$  but cannot resolve two features  $1\text{ }\mu\text{m}$  apart. Would increasing magnification to  $1000\times$  help? Why or why not?

Come to lecture to see the answer.

## Part V

# Telescopes

# Outline of Part V: Telescopes

The Refracting Telescope

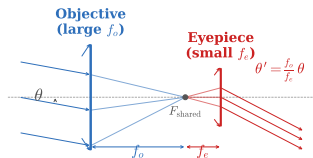
The Reflecting Telescope

# Outline of Part V: Telescopes

The Refracting Telescope

The Reflecting Telescope

# The Refracting Telescope



## Two Functions

- ▶ **Light gathering:** large objective collects more photons (brightness)
- ▶ **Angular magnification:** increases apparent angular size

For distant objects: the objective forms an image at its focal point. The eyepiece magnifies this image.

# Angular Magnification of a Telescope

## Telescope Magnification — (cp2e\_ch26\_s5)

$$M = \frac{\theta'}{\theta} = -\frac{f_o}{f_e}$$

$f_o$ : focal length of objective;  $f_e$ : focal length of eyepiece; negative sign = inverted image

### Design Principles

- ▶ Large  $f_o$  + small  $f_e \Rightarrow$  high magnification
- ▶ Large objective diameter  $\Rightarrow$  more light, better resolution
- ▶ Long tube length for large  $f_o$

### Telescope vs. Microscope

| Feature            | Telescope      | Microscope     |
|--------------------|----------------|----------------|
| Object dist.       | Very large     | Very small     |
| Priority Objective | Light collect. | Resolution     |
|                    | Gathers light  | Enlarges image |

# Outline of Part V: Telescopes

The Refracting Telescope

The Reflecting Telescope

# The Reflecting Telescope

Most large modern telescopes use a **concave mirror** as the objective — eliminating chromatic aberration and allowing very large apertures.

## Advantages over Refracting

- ▶ **No chromatic aberration:** mirrors reflect all wavelengths equally
- ▶ **Easier to build large:** mirrors can be supported from behind; lenses must be supported at edges
- ▶ **Hubble Space Telescope:** 2.4-m mirror
- ▶ **James Webb Space Telescope:** 6.5-m mirror

## Why does aperture matter?

Larger aperture  $\Rightarrow$  more photons collected per second  $\Rightarrow$  fainter objects visible.

Light-gathering power scales as  $D^2$ : doubling diameter quadruples light collected.

## Example: Telescope Magnification — (cp2e\_ch26\_ex6)

### Telescope Magnification

An astronomical telescope has an objective with  $f_o = 1.20$  m and an eyepiece with  $f_e = 40.0$  mm. Find the angular magnification.

$$M = -\frac{f_o}{f_e}$$

$$M = -\frac{1.20 \text{ m}}{0.0400 \text{ m}}$$

$$M = -30.0$$

The telescope magnifies angular size by  $30\times$ . The negative sign means the image is **inverted** (acceptable for astronomical use; terrestrial telescopes add an erecting lens).

- ▶ How would you increase  $M$  without changing  $f_o$ ?
- ▶ Why use a long  $f_o$  rather than a large lens power?

## Check Your Understanding

Two telescopes have the same magnification but different objective diameters. Which telescope can see fainter stars? Explain.

Come to lecture to see the answer.

## Part VI

# Aberrations

# Outline of Part VI: Aberrations

Types of Aberrations

# Outline of Part VI: Aberrations

Types of Aberrations

# Why Real Lenses Differ from Ideal Lenses

The thin-lens equation assumes **paraxial rays** (small angles) and a **perfectly spherical** lens of zero thickness. Real lenses violate these assumptions.

## Aberrations = Deviations from the Ideal

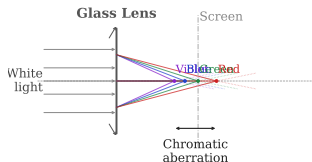
- ▶ **Chromatic aberration:** different wavelengths focus at different points
- ▶ **Spherical aberration:** marginal vs. paraxial rays focus at different points
- ▶ **Coma:** off-axis point sources produce comet-like blur
- ▶ **Astigmatism:** tangential vs. sagittal rays focus differently
- ▶ **Field curvature:** best-focus surface is curved, not flat

## Engineering tradeoff:

Correcting aberrations requires additional lens elements  $\Rightarrow$  more cost, weight, and complexity.

Perfect correction is never achieved — only minimized for the specific design purpose.

# Chromatic Aberration



## Cause

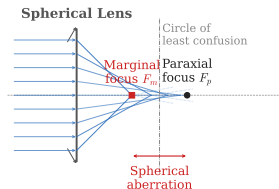
The index of refraction  $n(\lambda)$  varies with wavelength (**dispersion**).

$n$  larger for shorter  $\lambda$  (violet)  $\Rightarrow$  violet light bends more  $\Rightarrow$  shorter focal length

## Correction

- ▶ **Achromatic doublet:** crown glass + flint glass combine to equalize focal lengths for two wavelengths
- ▶ Reflecting telescopes: mirrors have no chromatic aberration

# Spherical Aberration



## Cause

A *spherical* surface does not focus all parallel rays to the same point.

Marginal (outer) rays refract too strongly; paraxial (central) rays refract correctly.

## Correction

- ▶ **Aspherical lens surfaces**
- ▶ **Stop down** (reduce aperture)  
— passes only paraxial rays
- ▶ **Multiple lens elements**  
cancel aberrations

## Other Aberrations and Correcting Strategies

### Coma

Off-axis point sources smear into comet-like tails. Gets worse at larger angles from the axis.

### Astigmatism

Rays in different planes focus at different distances. Not to be confused with the eye defect astigmatism (same principle).

### Field Curvature

The sharpest image lies on a curved surface, not a flat detector — problematic for cameras.

### Correction Strategies

- ▶ Use **multiple lens elements** (compound lens)
- ▶ Use **aperture stops** to restrict rays
- ▶ Use **aspherical surfaces**
- ▶ Use **mirrors** (chromatic: eliminated)
- ▶ Accept residual aberration appropriate for the application

Modern camera lenses use 10–20 elements to control all five Seidel aberrations simultaneously.

## Example: f-Number and Aperture — (cp2e\_ch26\_ex7)

### f-Number of a Camera Lens

A camera lens has a focal length of 50.0 mm and aperture diameter 25.0 mm. Find the f-number.

$$f/\# = \frac{f}{D}$$

$$f/\# = \frac{50.0 \text{ mm}}{25.0 \text{ mm}}$$

$$f/\# = 2.0$$

$f/2$  is a relatively “fast” (wide-aperture) lens — gathers much light, but spherical aberration is significant at full aperture.

Stopping down to  $f/8$  reduces aberrations but reduces light.

► Smaller  $f/\#$  means *larger* aperture (more light)

►  $M4 = 1/(8 \times f/\#)$

## Check Your Understanding

A refracting telescope shows a colored halo around a bright star. What type of aberration is this? How would switching to a reflecting telescope help?

Come to lecture to see the answer.

## Part VII

# Summary

# Outline of Part VII: Summary

Chapter Summary

# Outline of Part VII: Summary

Chapter Summary

# Chapter 26 Summary

## Key Equations

Eye power  $P = \frac{1}{d_o} + \frac{1}{d_i}$

Simple magnifier  $M = \frac{25 \text{ cm}}{f}$

Microscope  $m = m_o \cdot m_e$

Telescope  $M = -\frac{f_o}{f_e}$

Num. aperture  $NA = n \sin \alpha$

f-number  $f/\# = f/D$

## Conceptual Themes

- ▶ Eye fixes  $d_i$ ; adjusts  $P$  via accommodation
- ▶ Corrective lenses move virtual images to eye's usable range
- ▶ Color is a brain construction, not a physical property
- ▶ Magnification  $\neq$  resolution
- ▶ Aperture determines light gathering and resolution limit
- ▶ Aberrations are deviations from the paraxial ideal